Francesco Pio De Dominicis

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Turin, Italy

Education

Politecnico di Torino — Turin, Italy
M.Sc. in Computer Engineering - AI & Data Analysis — *2024 – Dec 2026 (expected)*
- **Thesis:** GroundOne: A Specialist Decision Model for Groundedness Verification in Local RAG.

Università degli Studi di Salerno — Salerno, Italy
B.Sc. in Computer Engineering (Final Grade: 109/110) — *2021 – 2024*
- **Thesis:** ROAM: cross-platform software for managing and controlling heterogeneous robot fleets.

Experience

AI Engineer — November 2026 – Present
Silicon Shoring FS Reply — *Turin, Italy*
- Bringing **AI agents** into every stage of the software development lifecycle for **financial services**.

AI Engineer - Intern — March 2026 – October 2026
Accenture — *Turin, Italy*
- Built **LangGraph** multi-agent workflows and enterprise **RAG** on **Azure OpenAI**, with evaluation and guardrails.

Software Engineer - Intern — June 2024 – August 2024
Youbiquo — *Cava de' Tirreni, Italy*
- Built a **Flutter** app integrated with **OpenRMF** for real-time teleoperation and telemetry of robot fleets.

Football Referee — 2017 – Present
A.I.A. - FIGC — *Campania & Piedmont, Italy*
- Split-second decisions under pressure and clear communication with assistant referees.

Projects

GroundOne (M.Sc. Thesis) | ***Python, MLX, Qwen2.5, LoRA, Swift, SwiftUI***
- Fully **local** specialist **Decision Model** that verifies, sentence by sentence and in a **single forward pass**, whether a RAG answer is supported by its sources, with no cloud APIs and no data leaving the device.
- Fine-tuned small **Qwen2.5** models (0.5B / 3B) with **LoRA** on **MLX** for **on-device** training and inference on Apple Silicon, delivered as a native **SwiftUI** app.

Mask Architecture Anomaly Segmentation for Road Scenes | ***Python, PyTorch***
- Developed an open-world semantic segmentation framework for autonomous driving using **Mask2Former** (EoMT) and a custom dynamic **Outlier Exposure** pipeline, documented in a paper-style project report.
- Implemented memory-efficient **Gradient Accumulation** to train and validate large Transformer architectures on resource-constrained hardware.

Hierarchical Coarse-to-Fine Concept Bottleneck Models | ***Python, PyTorch***
- Designed an **explainable** classifier built on a coarse-to-fine **Concept Bottleneck Model** with **LLM-generated concepts** and superclass gating, evaluated on **CIFAR-100** and documented in a paper-style project report.

Technical Skills

AI Engineering: LangGraph, LangChain, RAG, Hybrid Search (BM25 + dense), PGVector, FAISS, Azure OpenAI, Langfuse
ML & Local Inference: PyTorch, TensorFlow, scikit-learn, MLX, llama.cpp / GGUF, Ollama, Hugging Face, LoRA, Unsloth
Programming Languages: Python, Rust, Swift, Kotlin, Java, C, TypeScript, JavaScript, Dart, SQL
Backend & Data: FastAPI, Node.js, Spring Boot, Kafka, gRPC, PostgreSQL, MySQL, MongoDB, Redis, Apache Spark
DevOps & Cloud: Docker, Kubernetes, Helm, Argo CD, GitHub Actions, Terraform, Azure, Prometheus, Grafana
Languages: Italian (Native), English (C2 - Proficient)